CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 19: Constructions (Circles)

Every formula and result of the chapter on one sheet

1	Tangent at a point on a circle	Perpendicular to the radius at that point
2	Tangents from an external point P	Circle on OP as diameter cuts the given circle at the points of contact
3	Length of a tangent	$\sqrt{d^2 - r^2}$
4	Circumcentre	Meeting point of the perpendicular bisectors of the sides
5	Incentre	Meeting point of the bisectors of the angles
6	Circumradius	Distance from the circumcentre to a vertex
7	Inradius	Perpendicular distance from the incentre to a side
8	Circumcentre of a right-angled triangle	Mid-point of the hypotenuse; circumradius $= \frac{1}{2} \times$ hypotenuse
9	Equilateral triangle of side a	Circumradius $= \frac{a}{\sqrt{3}}$; inradius $= \frac{a}{2\sqrt{3}}$
10	Inradius of any triangle	$r = \frac{\text{area}}{s}$, where s is the semi-perimeter
11	Regular hexagon of side a	Circumradius $= a$; inradius $= \frac{\sqrt{3}}{2}a$